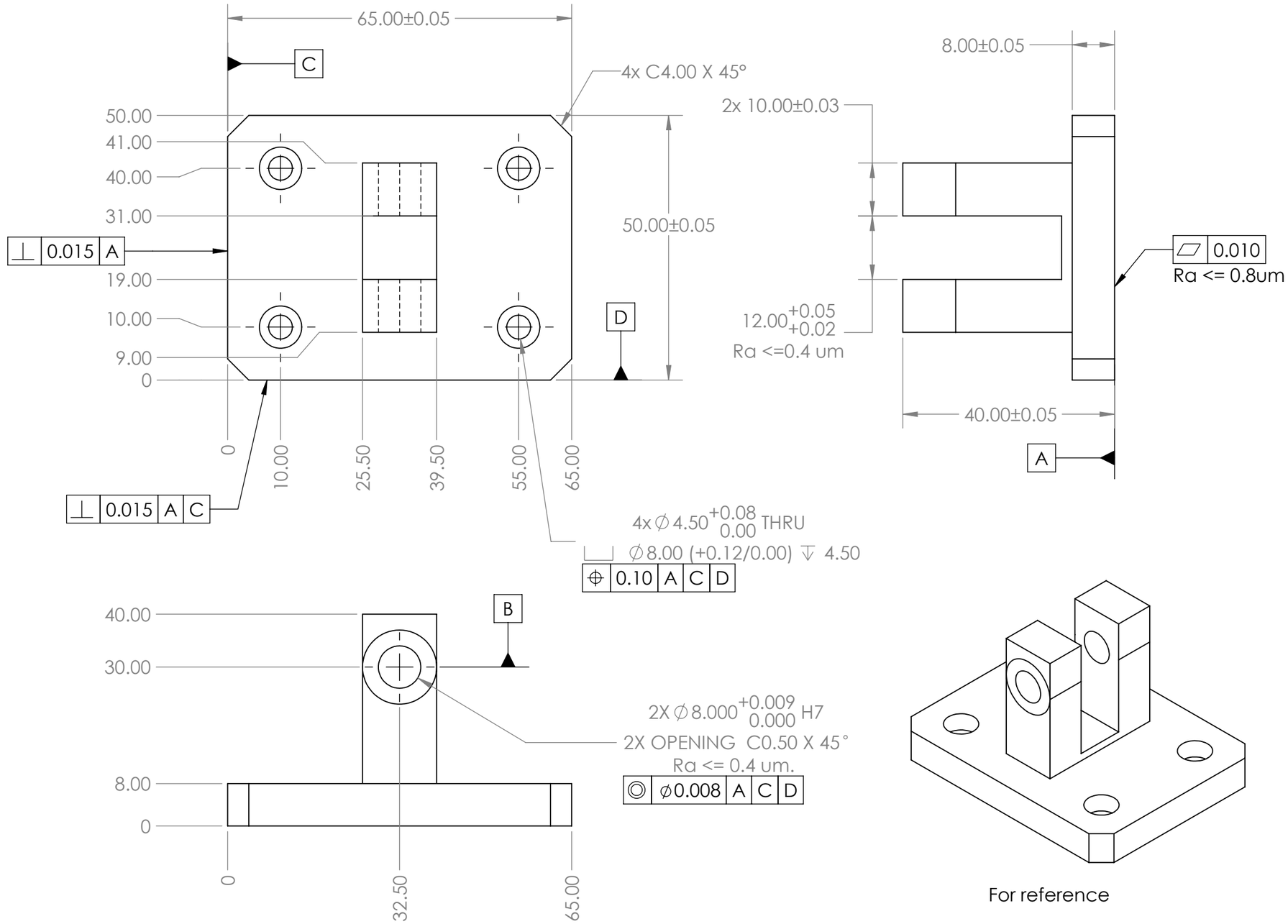
GENERAL NOTES: UNLESS OTHERWISE SPECIFIED

1. GENERAL & STANDARDS COMPLIANCE:
1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
1.2 3D CAD DATASET (W3-AERO-HINGE-BRACKET-2.4.STEP) GOVERNS NOMINAL GEOMETRY FOR UNTOLERANCED SURFACES.
1.3 GENERAL TOLERANCES: LINEAR +/-0.05 mm, ANGULAR +/-0.1 DEG (PRECISION AEROSPACE CNC MACHINING).
1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.
1.5 COORDINATE REFERENCE FRAME: FIRST OCTANT CONVENTION (ALL POSITIVE X, Y, Z FROM DATUM ORIGIN C-D-A).

2. MATERIAL & FLIGHT ASSURANCE:
2.1 MATERIAL OPTION A: AEROSPACE ALUMINUM-LITHIUM ALLOY 2195-T8 (AMS 4295), HIGH SPECIFIC STIFFNESS AND CRYOGENIC TOUGHNESS.
2.2 MATERIAL OPTION B: TITANIUM ALLOY Ti-6Al-4V GRADE 5 (AMS 4928 / ASTM B348), ANNEALED BILLET.
2.3 100% SUBTRACTIVE 5-AXIS CNC MACHINING ONLY. NO CASTING, WELDING, OR ADDITIVE MANUFACTURING PERMITTED TO ENSURE ULTRA-LOW OUTGASSING AND MICRO-CREEP RESISTANCE UNDER SPACE VACUUM.

3. STRUCTURAL LOADS & PIN ALIGNMENT CONTROLS:
3.1 QUASI-STATIC FLIGHT LOAD: WITHSTAND >= 25G LAUNCH ACCELERATION COMBINED LOAD WITHOUT PERMANENT PLASTIC SET (SAFETY FACTOR >= 1.50).
3.2 HINGE PIN COAXIALITY: TWIN CLEVIS HOLES (Ø8.000 +0.009/0.000 H7) MUST MAINTAIN COAXIALITY WITHIN Ø0.008 mm RELATIVE TO DATUM [B] TO GUARANTEE ULTRA-LOW DEPLOYMENT FRICTION TORQUE (DEPLOYMENT HYSTERESIS <= 0.05 mrad).
3.3 PRIMARY MOUNTING BASE [A]: FLATNESS <= 0.010 mm, SURFACE FINISH Ra <= 0.8 MICRONS TO PREVENT SATELLITE BULKHEAD PRELOAD DISTORTION.

4. THERMAL-VACUUM & SURFACE FINISH:
4.1 SURFACE FINISH: INTERNAL CLEVIS SLOT AND PIN BORES Ra <= 0.4 MICRONS. UNTOLERANCED SURFACES Ra <= 1.6 MICRONS.
4.2 VACUUM OUTGASSING REQUIREMENT: COMPLIANT WITH ECSS-Q-ST-70-02C (CVCM < 0.1%, TML < 1.0%).
4.3 DEBURRING: BREAK ALL SHARP EDGES 0.15-0.30 mm. ZERO BURRS PERMITTED TO AVOID SPACE DEBRIS / BINDING RISKS.



For reference

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: Linear ±0.05 mm, Angular ±0.1° (High-Precision 5-Axis CNC Milling)		FINISH: Chromic Acid Anodize (MIL-A-8625 Type I Class 1, for Al-Li) / Micro-Bead Blasted (for Ti)		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION: A
	NAME	SIGNATURE	DATE			TITLE: SPACECRAFT ULTRA-QUIET SOLAR ARRAY DEPLOYMENT HINGE BRACKET	
DRAWN							
CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL: Option A: Aerospace Al-Li Alloy 2195-T8 (AMS 4295) / Option B: Titanium Alloy Ti-6Al-4V Grade 5 (AMS 4928)		DWG NO. W3-AERO-HINGE-BRACKET-2.4	A3
				WEIGHT: ~89.2 g (Al-Li 2195) / ~145.8 g (Ti-6Al-4V)		SCALE : NTS	SHEET : 1 OF 1